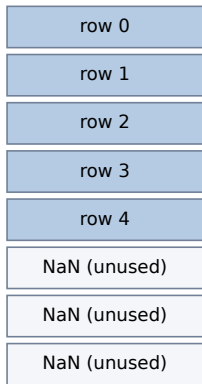


## Python ColumnBuffer

one (capacity × columns) float64 array

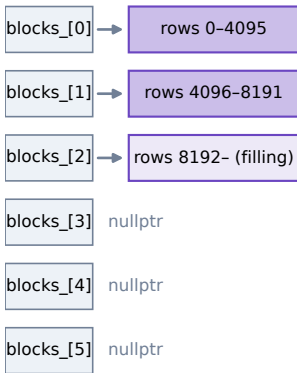


when full: allocate  $2 \times$  rows,  
copy the `_n` valid rows,  
swap the reference.  
A flush still holding the old  
array keeps reading valid rows.

`_n = 5`: incremented AFTER  
the row is written

## C++ `fr::Stream`

fixed pointer table + 4096-row blocks



`n_`: `std::atomic<size_t>`

writer: fill the row, then  
`n_.store(n+1, release)`

flush: `n_.load(acquire)`,  
read rows below it

blocks never move or  
get freed; at most  
65 536 blocks  
( $\approx 268$  M rows per stream)